

Hojoon Leo Kim

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RESEARCH INTERESTS

Machine Learning Systems, Hardware–Software Co-Design, Energy-Efficient AI

My research focuses on agent–hardware co-design: jointly optimizing agentic algorithms for planning and memory with accelerator architectures and execution mechanisms such as plan reuse. I extend this full-stack approach to embodied AI and robotics, where on-device vision–language–action (VLA) models require real-time, energy-efficient inference. My long-term goal is to develop reusable system abstractions and open benchmarks that bridge ML algorithms and accelerator architecture, enabling scalable deployment of agentic and embodied AI.

EDUCATION

- **Seoul National University** Seoul, South Korea
M.S. in Computer Science and Engineering Sep 2026 – Sep 2028 (Expected)
B.S. in Electrical and Computer Engineering, Summa Cum Laude Mar 2020 – Feb 2026
◦ GPA: 3.99/4.30 (major GPA: 4.10/4.30)
- **Stanford University** Jun 2025 – Aug 2025
Visiting student researcher Stanford, CA, USA
◦ GPA: 4.15/4.30; Courses: CS107 - Computer Organization and Systems (A+), CS161 - Design and Analysis of Algorithms (A)

PUBLICATIONS

C=CONFERENCE, S=IN SUBMISSION

* Equal contribution

- [S.4] Hojoon Kim, Chanyeong Lim, Jae W. Lee, and Kiwan Maeng. **Carnevale: Compiler-Architecture Co-Design for Efficient Masked Machine Learning at Scale**. *In submission to ASPLOS'27*
- [S.3] Jaehoon Yang, Yongbeom Kim, Hojoon Kim, Seung Yul Lee, and Jae W. Lee. **FLIT: A Fine-Grained Memory Management Layer for Colocating LLM Inference and Tuning**. *In submission to ATC'26*
- [S.2] Jongheon Jeong*, Hojoon Kim*, Rokhee Lee, Yeonhong Park, Young H. Oh, and Jae W. Lee. **Helios: Heterogeneous lightweight VLA model serving system**. *In submission to ASPLOS'27*
- [S.1] Seong Hoon Seo, Donghyun Lee, Geonha Lee, Hojoon Kim, Yeonhong Park, and Jae W. Lee. **QUESO: Storage-Assisted Quantization Error Compensation for On-Device LLM Inference**. *In submission to ATC'26*
- [C.3] Hojoon Kim, Yuheng Wu, and Thierry Tambe. **AgenticCache: Cache-Driven Asynchronous Planning for Embodied AI Agents**. *MLSys'26*. (Acceptance Rate: 26.4%) [Paper PDF]
- [C.2] Seung Yul Lee, Hojoon Kim, Yutack Park, Dawoon Jeong, Seungwu Han, Yeonhong Park, and Jae W. Lee. **FlashTP: Fused, Sparsity-Aware Tensor Product for Machine Learning Interatomic Potentials**. *ICML'25 Spotlight* (Spotlight Rate: 2.6%) [Paper PDF]
- [C.1] Yeonhong Park*, Jake Hyun*, Hojoon Kim, and Jae W. Lee. **DecDEC: A Systems Approach to Advancing Low-Bit LLM Quantization**. *OSDI'25*. (Acceptance Rate: 15.9%) [Paper PDF]

RESEARCH AND WORK EXPERIENCE

- **SNU Architecture and Code Optimization Lab [ARC Lab]** Seoul, South Korea
Graduate Researcher (P.I.: Professor Jae W. Lee) Feb 2026 – Aug 2028 (Expected)
Undergraduate Researcher Sep 2024 – Feb 2026
◦ Led the compiler–architecture co-design for exact INT64 arithmetic, developing optimized GPU kernels and proposing GPU microarchitectural extensions for scalable masked machine learning. (ASPLOS'27 submission)
◦ Led the design and implementation of HELIOS, a lightweight GPU kernel orchestrator for centralized multi-robot VLA serving. (ASPLOS'27 submission)
- **Stanford Tambe Lab [Tambe Lab]** Jun 2025 – Mar 2026
Undergraduate Researcher (P.I.: Professor Thierry Tambe) Stanford, CA
◦ Led research on cache-driven asynchronous planning for embodied AI agents by exploiting plan-level locality. (MLSys'26)
- **SNU Computer Architecture and Systems Lab [COMPARCH Lab]** Dec 2023 – Sep 2024
Research Intern (P.I.: Professor Jaewoong Sim) Seoul, South Korea
◦ Authored undergraduate thesis based on **Tender (ISCA'24)** quantization architecture; designed hardware module to shift offline computation to runtime, improving model accuracy.
- **MODULABS [MODULABS]** Jun 2021 – Sep 2021
Backend Software Engineer Seoul, South Korea

ACADEMIC PROJECTS

- **A File System with Full-Path Indexing: Operating Systems (1st Place)** Dec 2024
Reduced file lookup latency from 24,000 to 993 cycles on the xv6-riscv kernel
- **Pipelined Central Processing Unit with Perceptron Branch Predictor: Computer Organization** Jun 2024
- **L1, L2, and Memory Three-Level Memory Simulator with MSHR: Computer Organization** Jun 2024
- **A CNN Accelerator on FPGA (Artix-7): Digital Systems Design and Lab (1st Place)** Dec 2023

HONORS AND AWARDS

- **SNU Semiconductor Specialized Scholarship (KRW 15,000,000)** Jan 2024 – Feb 2026
SNU Semiconductor Specialization Project Group
- **Grand Prize (1st Place), 2025 AI Chip Contest - NPU Optimization Track (KRW 10,000,000)** Nov 2025
Ministry of Science and ICT (MSIT), Republic of Korea
- **Global Leadership Program (airfare)** Jun 2025 – Aug 2025
College of Engineering
- **Merit-Based Tuition Scholarship (70% of tuition)** Aug 2023 – Jun 2025
Department of Electrical and Computer Engineering
- **1st Prize – Advanced CUDA Accelerator Programming School** Feb 2025
SNU Thunder Research Group
- **4th Prize – Military AI Contest (KRW 2,000,000)** Dec 2022
Ministry of National Defense & Ministry of Science and ICT

TEACHING EXPERIENCE

- **Peer Tutor, Computer Organization; Programming Methodology** Sep 2024 – Jun 2025
Department of Electrical and Computer Engineering

EXTRACURRICULAR ACTIVITIES

- **Member, SNU Business School Tennis Club** Mar 2024 – Dec 2024
- **Peer Mentor, SNU Center for Campus Life & Culture** Mar 2024 – Jun 2024
 - Received the Best Mentor Award and Best Mentoring Team Award.
- **SIGINT Specialist, Republic of Korea Army** Jan 2022 – Jul 2023
 - Two-time recipient of the Best SIGINT Analyst Award; recognized as a First-Class Soldier (fitness, mental readiness, and specialty proficiency).

RELEVANT COURSES

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| • System Programming (Prof. Jae W. Lee) | • Advanced Compiler (Prof. SooMook Moon) |
| • Operating Systems (Prof. JinSoo Kim) | • Computer Networks (Prof. KyoungSoo Park) |
| • Computer Organization (Prof. Jaewoong Sim) | • Linear Algebra (Prof. Young-Hoon Kiem) |
| • Advanced Computer Architecture (Prof. Sungjoo Yoo) | • Parallel Distributed Computing (Prof. Jinho Lee) |
| • Digital Systems Design (Prof. Jaewoong Sim) | • Computer Organization & Design (Prof. Jangwoo Kim) |

TECHNICAL STRENGTHS

- **Computer Languages:** C/C++, Verilog/SystemVerilog, Python
- **Frameworks and Skills:** CUDA, PyTorch, Nsight System/Compute

LANGUAGE PROFICIENCY

Korean (Native), English (TOEFL 106 [R:30/L:28/S:22/W:26], OPIc AL)